

Performance Benchmark Report

Sports Video Ingestion and Preprocessing Pipeline for Computer Vision Applications

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Overview

This report summarizes the benchmark setting used to evaluate the sports video ingestion and preprocessing pipeline. The purpose of the benchmark is to document the computing environment and the input video characteristics used for testing the pipeline under different video durations and resolutions. The pipeline accepts raw sports videos, extracts metadata, samples frames, normalizes frame resolution, converts frames into RGB and grayscale outputs, computes frame-quality scores, detects scene boundaries, and writes structured manifests and validation reports. The generated RGB frame folders can then be used as input for downstream computer vision tasks such as object detection or pose estimation.

Four sports videos were used in the benchmark: American football, basketball, rugby, and beach volleyball. These videos provide variation in sport type, motion pattern, frame rate, total duration, total number of frames, and spatial resolution. The selected set includes both short and long videos and covers two spatial-resolution conditions: 1280×720 and 1920×1080 . This satisfies the requirement of benchmarking across at least two video durations and two video resolutions.

System Environment

The benchmark was performed on an Ubuntu workstation. The system used a 12-core / 24-thread Intel Core i9 CPU, 125 GiB system memory, and three NVIDIA GPUs. The GPU configuration included two RTX 3050 cards and one GTX 1080 Ti card. FFmpeg and FFprobe were used for video handling and metadata extraction. The active Python environment was a Conda environment named `sam2`.

Benchmark Video Set

The benchmark used four real sports videos. Table 2 summarizes the input characteristics. The American football and basketball videos represent longer-duration inputs, while rugby and beach volleyball represent shorter-duration inputs. American football, basketball, and rugby were 720p inputs, while beach volleyball was a 1080p input.

Processing Behaviour

The video set was chosen to evaluate how the preprocessing pipeline behaves under different temporal and spatial conditions. Longer videos contain more frames and therefore require more decoding, sampling, frame-quality scoring, scene-difference computation, and output writing. Higher-resolution videos contain more pixels per frame and therefore increase the cost of resizing,

Table 1: Benchmark system environment.

Component	Specification
Operating system	Ubuntu 22.04.5 LTS
Kernel	Linux 6.8.0-111-generic
CPU	Intel Core i9-7920X, 12 cores / 24 threads, 2.90 GHz
System memory	125 GiB RAM
GPU 1	NVIDIA GeForce RTX 3050, 8 GiB
GPU 2	NVIDIA GeForce GTX 1080 Ti, 11 GiB
GPU 3	NVIDIA GeForce RTX 3050, 8 GiB
NVIDIA driver	580.142
CUDA runtime reported by <code>nvidia-smi</code>	13.0
Python environment	Conda environment <code>sam2</code> , Python 3.12.4
FFmpeg / FFprobe	4.4.2

Table 2: Benchmark video inputs.

Sport	FPS	Resolution	Total Frames	Duration
American football	59.94	1280×720	16,340	272.61 s
Basketball	29.97	1280×720	3,323	110.88 s
Rugby	30.00	1280×720	474	15.80 s
Beach volleyball	23.98	1920×1080	285	11.89 s

color conversion, and image-quality analysis. In this benchmark, the American football video provides the longest and highest-frame-count case, while the beach volleyball video provides the 1080p high-resolution case.

The preprocessing workload is mainly CPU and I/O oriented because it involves video decoding, frame transformation, image writing, JSON/JSONL report generation, and validation. GPU utilization is expected to remain low during preprocessing unless downstream deep-learning inference is executed in the same run. This separation is intentional: the benchmark focuses on the ingestion and preprocessing stage, while the generated RGB frame outputs are prepared for later downstream computer vision models.

Summary

The benchmark setup covers four sports videos across two duration groups and two resolution conditions. The videos include American football, basketball, rugby, and beach volleyball. The environment table documents the CPU, GPU, memory, operating system, Python environment, and FFmpeg/FFprobe versions used for the run. The input table documents the video duration, FPS, resolution, and frame count. Together, these provide a concise performance benchmark context for evaluating the sports video ingestion and preprocessing pipeline.